Model C Fit to the Real Data

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1 Summary

Model C (hierarchical over sessions, one group per hand, distractor amplitude as a covariate) was fit to all 12,937 trials of the Sirius data (42 sessions). The main results:

- **The PSE shift grows with distractor amplitude.** The hand-level distractor coefficient is clearly positive, $\mu_{\beta_2} = 0.90$ [0.71, 1.09] (left) and 1.18 [0.99, 1.37] (right). For a typical session the bimanual PSE lies 1.2 (left) and 1.7 (right) μm below the unimanual one with a 6 μm distractor, and 6.6 and 8.9 μm below with 32 μm. Averaged over the 42 sessions the shift is -4.5 μm (left) and -5.7 μm (right), close to Model B's estimates.
- **The fit is reliable but the model misfits in specific places.** The sampler converged without problems. However, the bimanual trials at the weakest stimuli contain more "high" responses than Model C predicts, and LOO prefers Model B by 37.1 (SE 11.9), with most of the difference at the 6 and 12 μm stimuli.
- **The misfit follows from three assumptions of Model C**: unimanual and bimanual trials share their lapse rates, they share their slope, and the PSE shift is proportional to the distractor amplitude, so it shrinks to 0 for a weak distractor. For the right hand, the 6 μm sessions shift by 3.9 μm on average under Model B but only 1.7 μm under Model C.
- The amplitude dependence itself is robust, but its size at small amplitudes, and the shared lapse rates and slopes, should be revisited before Model C's estimates are used for inference (Section 6).

Every figure and number in this report is produced by `Main_Analysis_C.py`; the cell "Numbers for Fit_C_Report.tex" prints the numbers that the earlier cells do not. All intervals are 95% highest density intervals (HDIs). PSE and JND are in micrometres of stimulus amplitude: $\text{PSE}_{\mu m} = 28 + 14.63\,\text{PSE}$ and $\text{JND}_{\mu m} = 14.63\,\text{JND}$, where 28 and 14.63 are the mean and sd of the 8 amplitude levels.

2 Model and Fit

The model is the one described in the Model C convergence report. For hand j and session s,

$$\psi(x, d) = \gamma_{h,js} + (1 - \gamma_{h,js} - \gamma_{l,js})\ \text{logit}^{-1}\left(\beta_{0,js} + \beta_{1,js}\, x + \beta_{2,js}\, d\right),$$

where x is the standardized stimulus amplitude and d the distractor amplitude divided by 14.63 ($d = 0$ on unimanual trials; every session uses a single distractor amplitude on its bimanual trials).

The five session parameters are drawn from hand-level normal (for β) and logit-normal (for γ) distributions. The priors on β_0, β_1 and the lapse rates are those of Model B, and $\mu_{\beta_2} \sim \mathcal{N}(0, 1)$, $\sigma_{\beta_2} \sim \text{HalfNormal}(1)$.

Because $\beta_2 d$ only shifts the curve, the PSE shift of a session is $-\beta_2 d_s/\beta_1$ standardized units, and its JND and lapse rates are the same in both conditions.

The fit used NUTS with 4 chains of 1000 tuning and 5000 kept draws (20,000 draws in total) and took 9 minutes. The data have 3,230–3,237 trials in each hand × condition cell.

2.1 Sampling diagnostics

There were 2 divergences in the 20,000 draws, and the tree depth never exceeded 5. All 1,300 variables, including the raw sampler variables, have $\hat{R} = 1.000$, bulk ESS $\geq$ 3,410 and tail ESS $\geq$ 5,231. Unlike Model B, whose raw `z_sig_b0` variables showed $\hat{R}$ warnings from an unidentified sign, Model C defines its HalfNormal sds directly, so no such warning appears.

Figures 1–3 show trace plots for a few parameters: the left hand's session parameters in session 04-13 (12 μm distractor; the session whose curves are shown below) and 05-12 (32 μm), the hyperparameters of β_2, and the lapse rate γ_h with its hyperparameters. The four chains overlap and mix quickly, including for σ_{γ_h}, which piles up against 0. The gray ticks on the axes mark the 2 divergent draws; they lie in no particular region of the posterior.

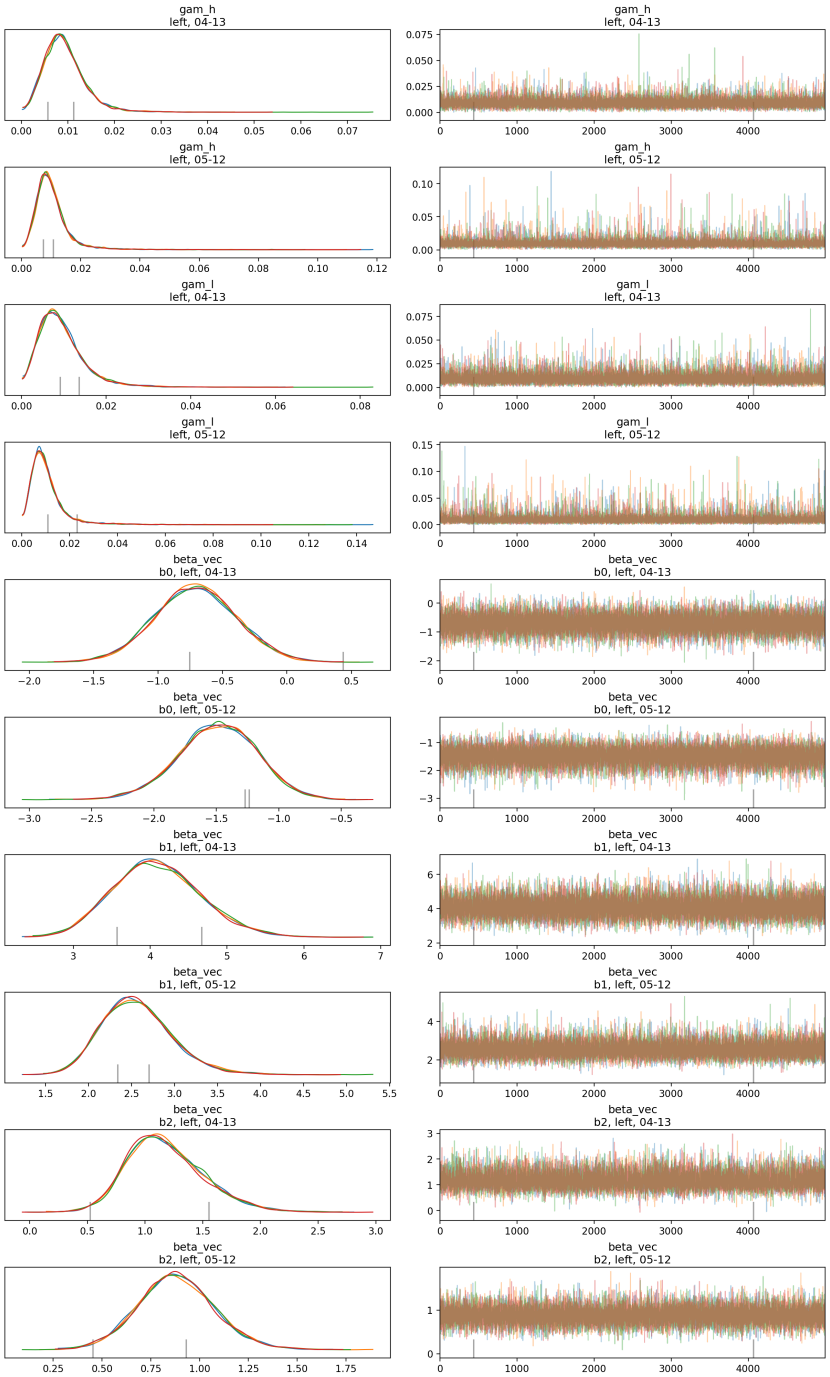


Figure 1: Trace plots (left: density per chain; right: draws per chain) of the left hand's session-level γ_h, γ_l, β_0, β_1 and β_2 in sessions 04-13 and 05-12.

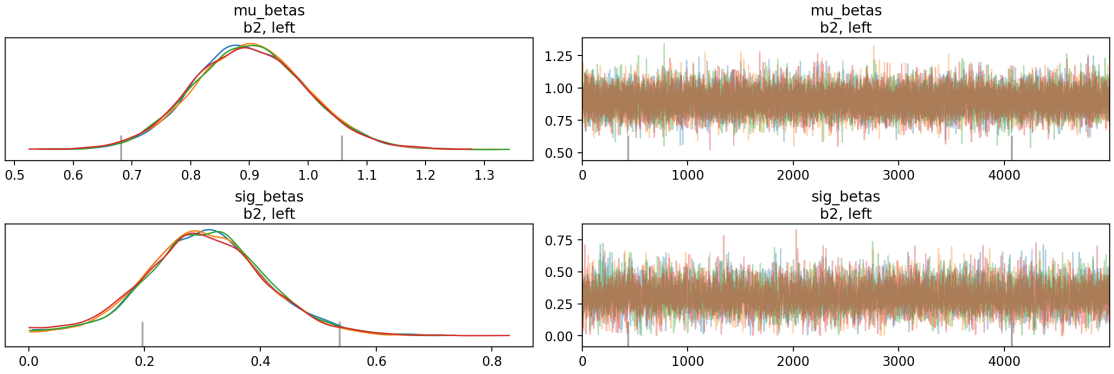


Figure 2: Trace plots of the left hand's μ_{β_2} and σ_{β_2}.

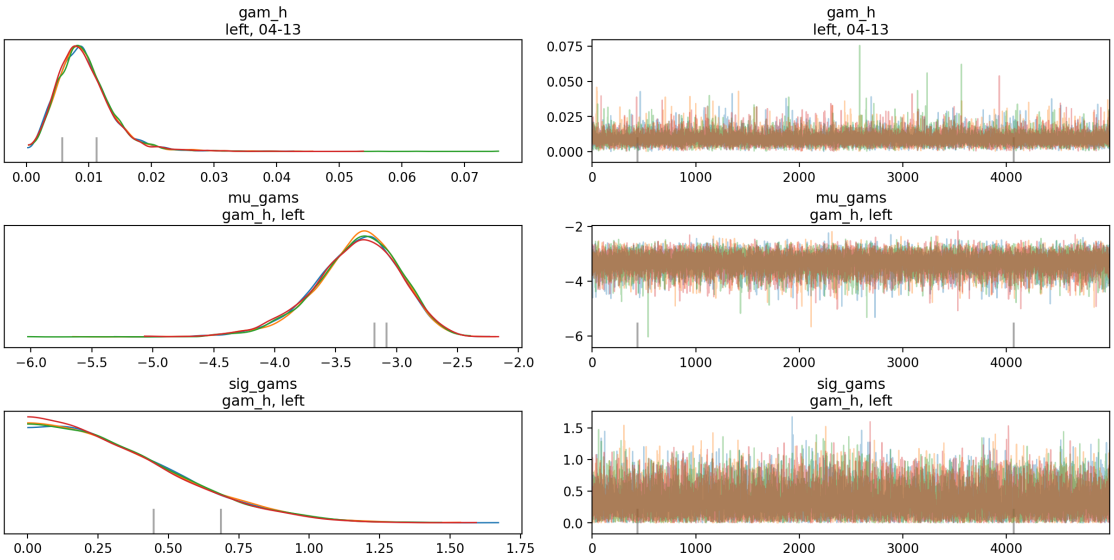


Figure 3: Trace plots of the left hand's γ_h on 04-13 and of the hyperparameters μ_{γ_h} and σ_{γ_h} it is drawn from.

2.2 Posterior predictive check

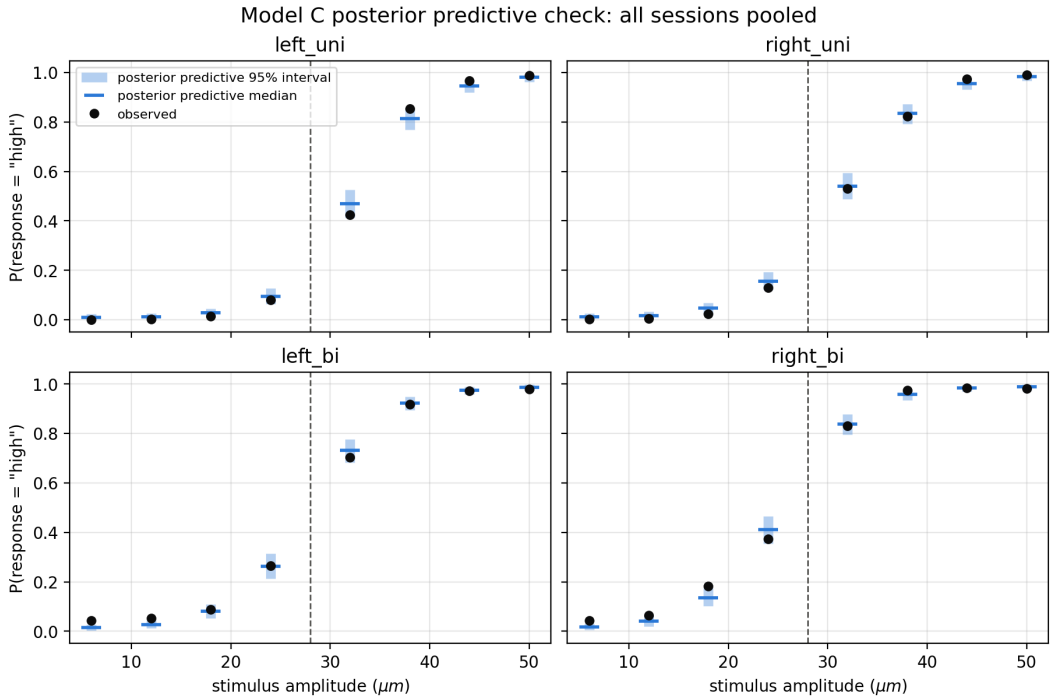


Figure 4: Observed proportion of "high" responses at each amplitude, pooled over sessions (black), against the posterior predictive median and 95% interval (blue), by hand and condition. The dashed line is the trained threshold of 28 μm.

Of the 32 hand × condition × amplitude proportions, 26 lie inside their 95% posterior predictive intervals (Figure 4). The six outside are all at the weak end of the stimulus range (Table 1): on bimanual trials Model C predicts too few "high" responses at 6–18 μm, and on left unimanual trials at 6 μm it predicts a few where the monkey gave none.

Table 1: Proportions outside their 95% posterior predictive interval.

hand	condition	amplitude (μm)	observed	predicted median	95% interval
left	unimanual	6	0.000	0.010	[0.002, 0.025]
left	bimanual	6	0.042	0.015	[0.002, 0.027]
left	bimanual	12	0.052	0.028	[0.012, 0.045]
right	bimanual	6	0.042	0.017	[0.005, 0.032]
right	bimanual	12	0.065	0.040	[0.020, 0.062]
right	bimanual	18	0.183	0.136	[0.101, 0.175]

This is the pattern expected if the distractor raises the lapse rate γ_h but the model can only use one lapse rate for both conditions. The fitted typical lapse rates ($\gamma_h = 0.009$ and 0.011, Table 3) lie between the near-zero unimanual and the higher bimanual rates that Model B estimates separately: too high for unimanual trials and too low for bimanual ones.

Within single sessions. Each session × hand × condition × amplitude cell holds about 10 trials. Of the 1,344 cells, 1,331 (99.0%) have their observed count of "high" inside the 95% posterior predictive interval, and 2.0% and 1.7% of the PIT values are below 0.025 and above 0.975, against 2.5% each for a calibrated model (Figure 5). Each cell's interval comes from a model fit partly to that same cell, so this is a check for gross misfit, and none appears: the lowest coverage, 0.93, is for right-hand bimanual trials at 6 μm. The misfit of Table 1 is systematic but small in any one session, and only shows when the sessions are pooled.

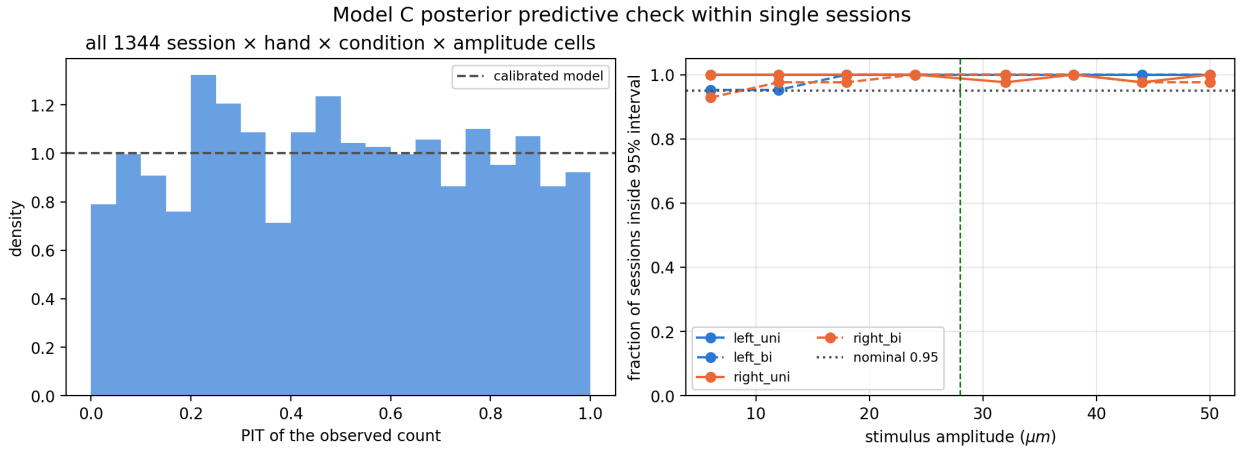


Figure 5: Posterior predictive check within single sessions. Left: PIT values of the observed number of "high" responses in each session × hand × condition × amplitude cell (flat at 1 for a calibrated model). Right: fraction of the 42 sessions whose observed count lies inside its 95% predictive interval.

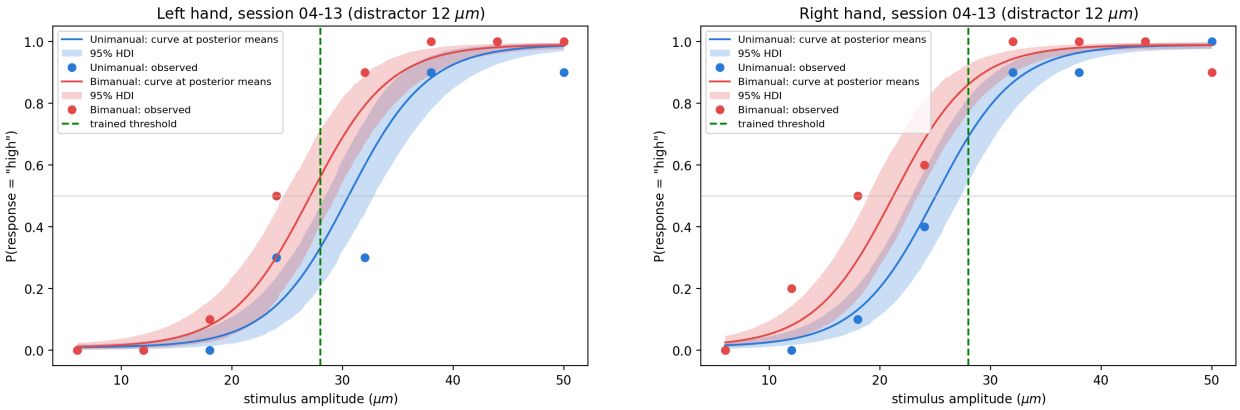


Figure 6: Fitted psychometric curves for session 04-13 (12 μm distractor): the curve at the posterior means of the parameters, the pointwise 95% HDI of the curve, and the observed proportions of "high" responses (9–10 trials per point). Unimanual in blue, bimanual in red; the dashed line is the trained threshold.

Figure 6 shows the fit in one session, 04-13, the session shown in the Model B report. The two curves of a hand have the same slope and asymptotes and differ only in position. The bimanual curve lies to the left of the unimanual one, with session PSEs of 30.6 (unimanual) and 27.1 (bimanual) μm

for the left hand and 25.0 and 21.2 μm for the right. With about 10 trials per point the observed proportions scatter around the curves; single stray points, such as the left unimanual proportion of 0.3 at 32 μm, pull the curve only partly towards them.

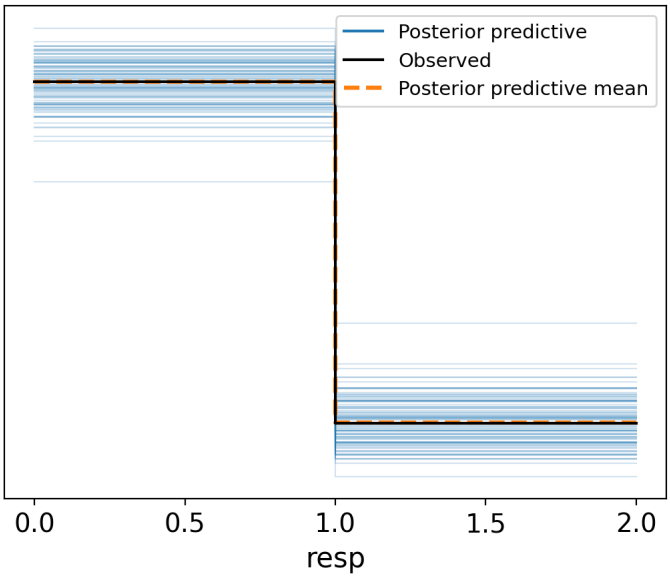


Figure 7: ArviZ posterior predictive check of the responses (0 = "low", 1 = "high"), pooled over all trials: observed (black), 100 posterior predictive draws (blue) and their mean (orange dashed).

The overall proportion of "high" responses is reproduced (Figure 7); for binary data this check is coarse, and Figure 4 is the more informative one.

3 Population-level Results

Table 2: Hand-level hyperparameters: posterior mean [95% HDI]. The lapse-rate hyperparameters are on the logit scale.

	left	right
μ_{β_0}	−1.30 [−1.56, −1.04]	−0.92 [−1.29, −0.55]
μ_{β_1}	4.39 [3.95, 4.85]	4.25 [3.92, 4.62]
μ_{β_2}	0.90 [0.71, 1.09]	1.18 [0.99, 1.37]
σ_{β_0}	0.60 [0.39, 0.81]	1.07 [0.79, 1.36]
σ_{β_1}	0.90 [0.57, 1.26]	0.35 [0.00, 0.69]
σ_{β_2}	0.30 [0.08, 0.51]	0.23 [0.00, 0.45]
μ_{γ_h}	−3.33 [−4.07, −2.69]	−3.11 [−3.91, −2.43]
μ_{γ_l}	−3.42 [−4.36, −2.60]	−3.41 [−4.18, −2.67]
σ_{γ_h}	0.34 [0.00, 0.82]	0.34 [0.00, 0.82]
σ_{γ_l}	0.40 [0.00, 0.96]	0.47 [0.00, 1.03]

Table 3: A typical session (all parameters at their hand-level means): unimanual PSE, JND and lapse rates, posterior mean [95% HDI]. The JND and lapse rates are the same on bimanual trials in Model C.

	left	right
PSE, unimanual (μm)	32.34 [31.51, 33.17]	31.15 [29.92, 32.40]
JND (μm)	3.75 [3.39, 4.11]	3.88 [3.59, 4.19]
γ_h	0.0091 [0.0033, 0.0147]	0.0113 [0.0042, 0.0189]
γ_l	0.0086 [0.0023, 0.0155]	0.0085 [0.0029, 0.0143]

The unimanual PSE of a typical session lies 4.3 (left) and 3.2 (right) μm above the trained threshold of 28 μm (Table 3). The sessions differ in their bias ($\sigma_{\beta_0} = 0.60$ and 1.07) and, for the left hand, in their slope ($\sigma_{\beta_1} = 0.90$): the left-hand session JNDs range from 2.95 to 6.59 μm, the right-hand ones only from 3.65 to 4.31 μm. As in Model B and in Model C's convergence study, σ_{γ_h} and σ_{γ_l} are not identified; their posteriors largely reflect the prior.

The data determine all five typical-session quantities rather than the priors: the posterior sd is 4.5–6.6% of the prior sd for the unimanual PSE, 5.9–7.3% for the JND, 9–12% for the lapse rates and 9.6–9.7% for μ_{β_2} (Figures 8–12).

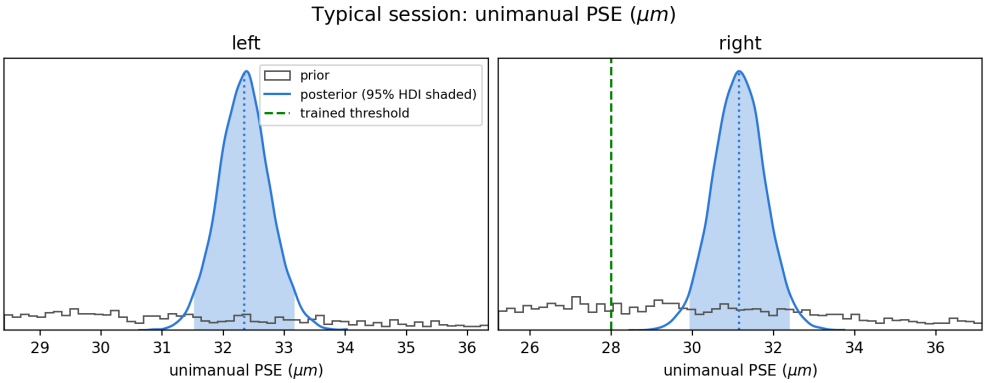


Figure 8: Prior (gray) and posterior (blue) of a typical session's unimanual PSE.

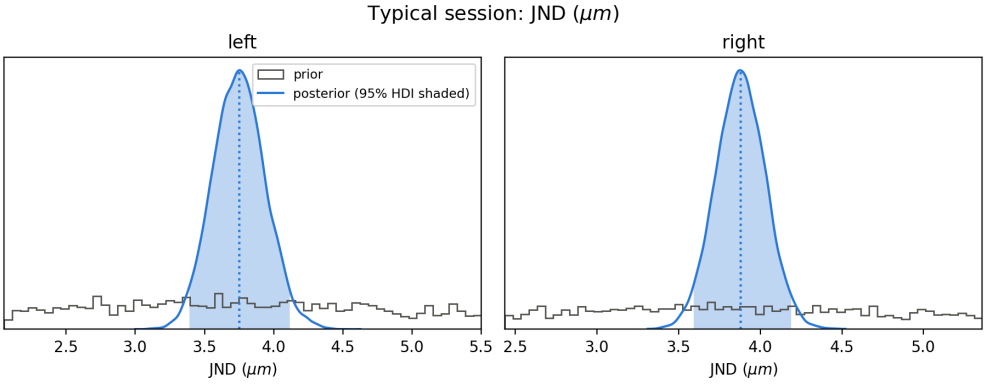


Figure 9: Prior (gray) and posterior (blue) of a typical session's JND, the same in both conditions.

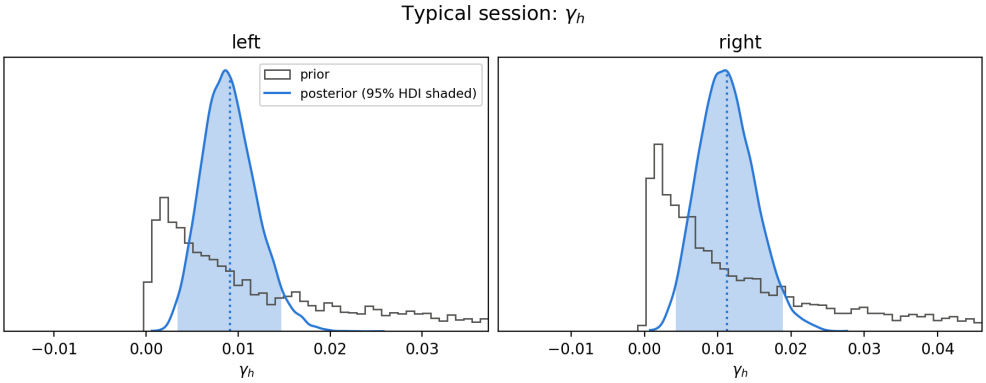


Figure 10: Prior (gray) and posterior (blue) of a typical session's γ_h.

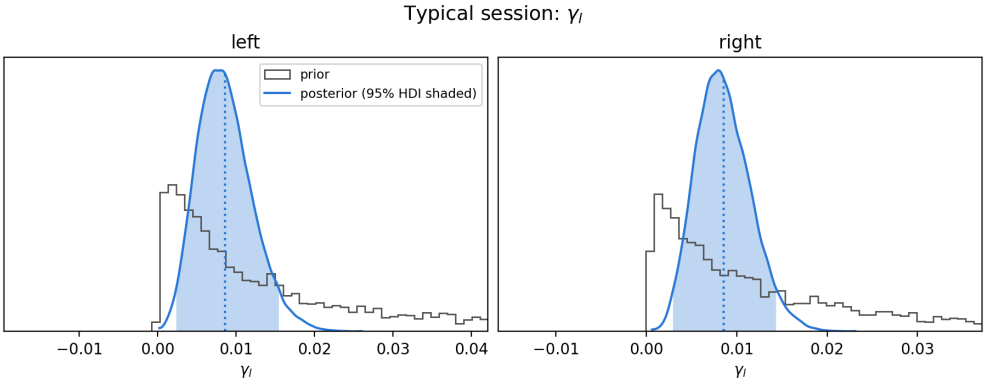


Figure 11: Prior (gray) and posterior (blue) of a typical session's γ_l.

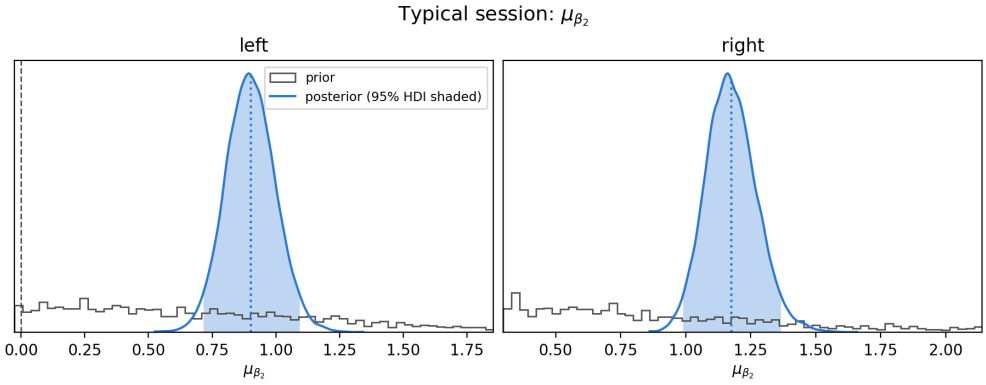


Figure 12: Prior (gray) and posterior (blue) of the hand-level distractor coefficient μ_{β_2}; the dashed line marks 0.

4 Effect of the Distractor

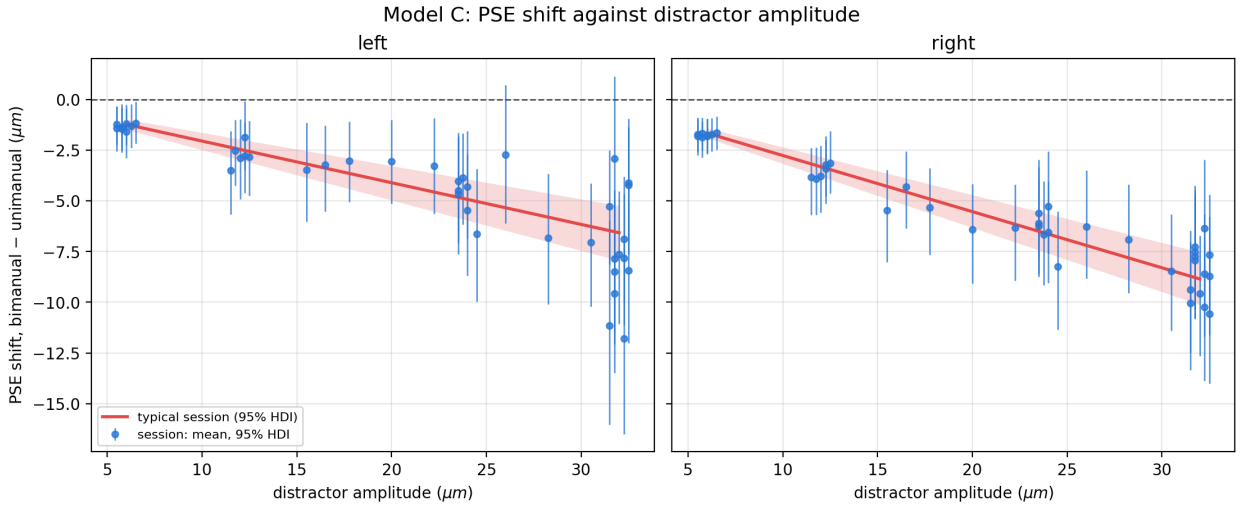


Figure 13: PSE shift (bimanual − unimanual) against the session's distractor amplitude. Blue: every session, posterior mean and 95% HDI (sessions sharing an amplitude are spread slightly). Red: a typical session, posterior mean and 95% HDI.

Table 4: PSE shift (bimanual − unimanual, μm) of a typical session at the most used distractor amplitudes: posterior mean [95% HDI].

distractor (μm)	left	right
6	−1.23 [−1.49, −0.98]	−1.66 [−1.90, −1.41]
12	−2.46 [−2.98, −1.97]	−3.32 [−3.79, −2.83]
24	−4.93 [−5.96, −3.93]	−6.64 [−7.59, −5.65]
32	−6.57 [−7.95, −5.24]	−8.85 [−10.12, −7.54]

The distractor lowers the PSE, and by more the stronger it is (Figure 13, Table 4). The shift is larger for the right hand at every amplitude. Averaged over the 42 sessions it is −4.51 [−5.15, −3.86] μm for the left hand and −5.74 [−6.39, −5.11] μm for the right. At the session level, the shift's HDI lies below 0 in 40 of 42 sessions (left) and all 42 (right).

The sessions vary only a little around the hand-level line: σ_{β_2} is 0.30 (left) and 0.23 (right), so the spread of the session shifts in Figure 13 mostly reflects the sessions' different distractor amplitudes rather than different β_2. The convergence study showed that a session's β_2 is poorly determined when its distractor is weak and is then pulled toward the hand mean; the 6 μm sessions in Figure 13 sit almost exactly on the line.

4.1 Does the shift scale with amplitude?

Model C assumes the shift is proportional to the distractor amplitude. A direct check needs a measure of the shift that does not depend on the model: for the sessions at each of the four most used amplitudes, the difference between the proportions of "high" responses on bimanual and unimanual trials, over all stimuli. It reflects the PSE shift and, to a lesser extent, the higher lapse

rate γ_h on bimanual trials. Table 5 compares the observed difference with its posterior predictive distribution; Figure 14 shows the proportions behind it.

Table 5: Bimanual − unimanual difference in the proportion of "high" responses, in the sessions at each distractor amplitude: observed, and posterior predictive median and 95% interval. P: posterior predictive probability of a difference below the observed one.

hand	distractor	sessions	observed	predicted	95% interval	P
left	6 μm	8	0.011	0.035	[0.002, 0.068]	0.07
left	12 μm	6	0.081	0.050	[0.014, 0.087]	0.95
left	24 μm	7	0.092	0.082	[0.046, 0.119]	0.71
left	32 μm	13	0.148	0.131	[0.099, 0.164]	0.85
right	6 μm	8	0.073	0.040	[0.009, 0.071]	0.98
right	12 μm	6	0.088	0.063	[0.027, 0.103]	0.90
right	24 μm	7	0.113	0.114	[0.076, 0.151]	0.49
right	32 μm	13	0.161	0.153	[0.122, 0.184]	0.69

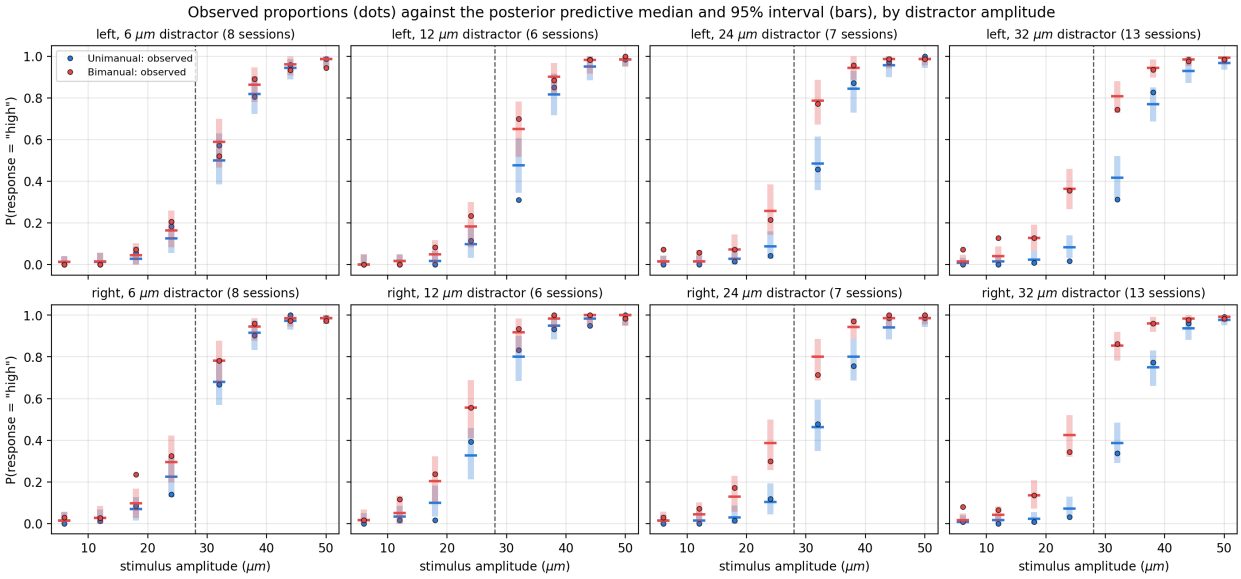


Figure 14: Observed proportions of "high" responses (dots) against the posterior predictive median and 95% interval (bars), in the sessions at each distractor amplitude. Unimanual in blue, bimanual in red.

Seven of the eight observed differences lie inside their predictive intervals. The exception is the right hand at 6 μm: the observed difference, 0.073, lies above the interval [0.009, 0.071], and the model predicts a smaller difference in 98% of its replicates. In Figure 14 this shows at 18 μm, where the right-hand bimanual proportion lies well above its predicted range. For the right hand, then, even a weak distractor shifts responses more than a proportional model allows, the presence effect suggested by the comparison with Model B. The left hand at 6 μm goes the other way: its observed difference is smaller than predicted ($P = 0.07$), so for the left hand a proportional shift is, if anything, too large at weak distractors.

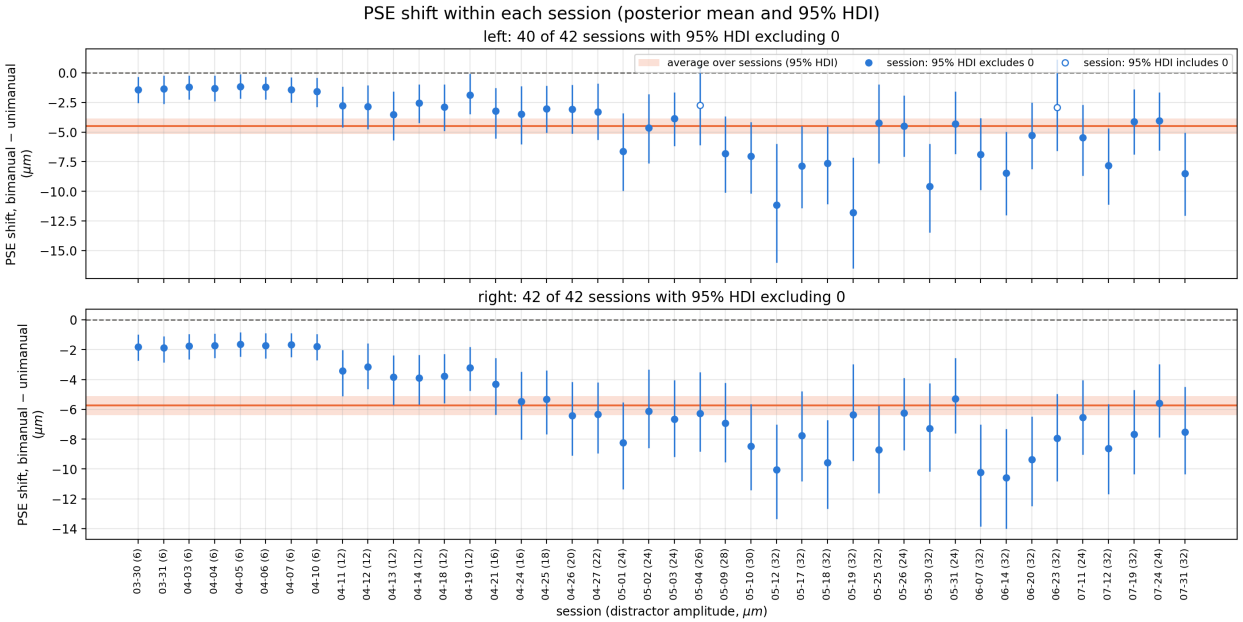


Figure 15: PSE shift (bimanual − unimanual) within each session, in date order with the distractor amplitude in brackets: posterior mean and 95% HDI (filled: HDI excludes 0). The orange band is the average over the 42 sessions.

In date order (Figure 15) the session shifts follow the amplitude schedule: small in the first eight sessions, which all used 6 μm, and largest in the 32 μm sessions from May onwards.

By construction, Model C cannot show an effect of the distractor on the JND or the lapse rates.

4.2 Session estimates

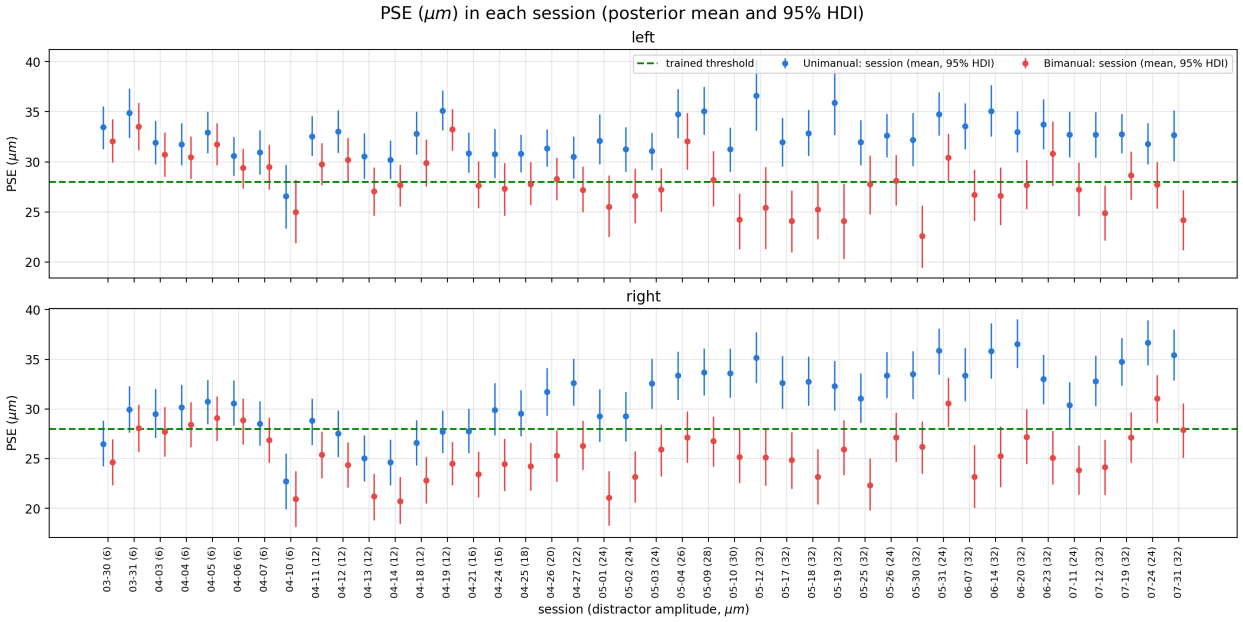


Figure 16: Unimanual (blue) and bimanual (red) PSE in each session: posterior mean and 95% HDI. The dashed line is the trained threshold.

The session PSEs range from 26.6 to 36.6 μm (left) and 22.7 to 36.7 μm (right) on unimanual trials, and from 22.6 to 33.5 and 20.7 to 31.1 μm on bimanual trials (Figure 16). Both conditions move together from session to session, which the within-session shift cancels.

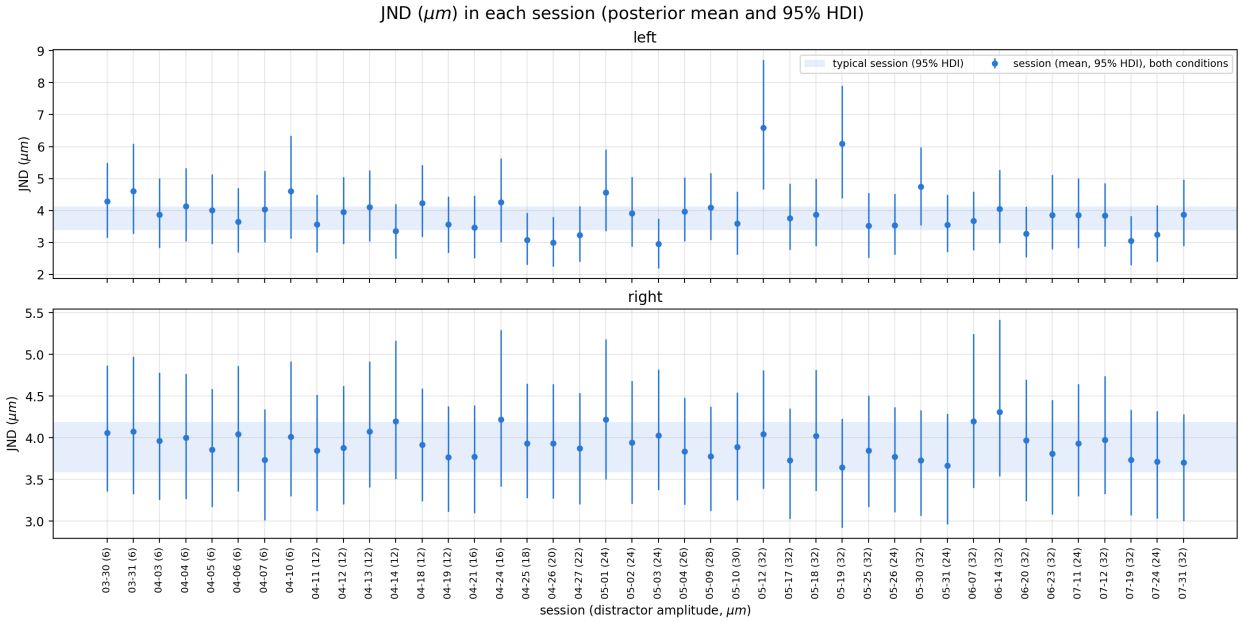


Figure 17: JND in each session, shared by both conditions: posterior mean and 95% HDI. The band is a typical session's 95% HDI.

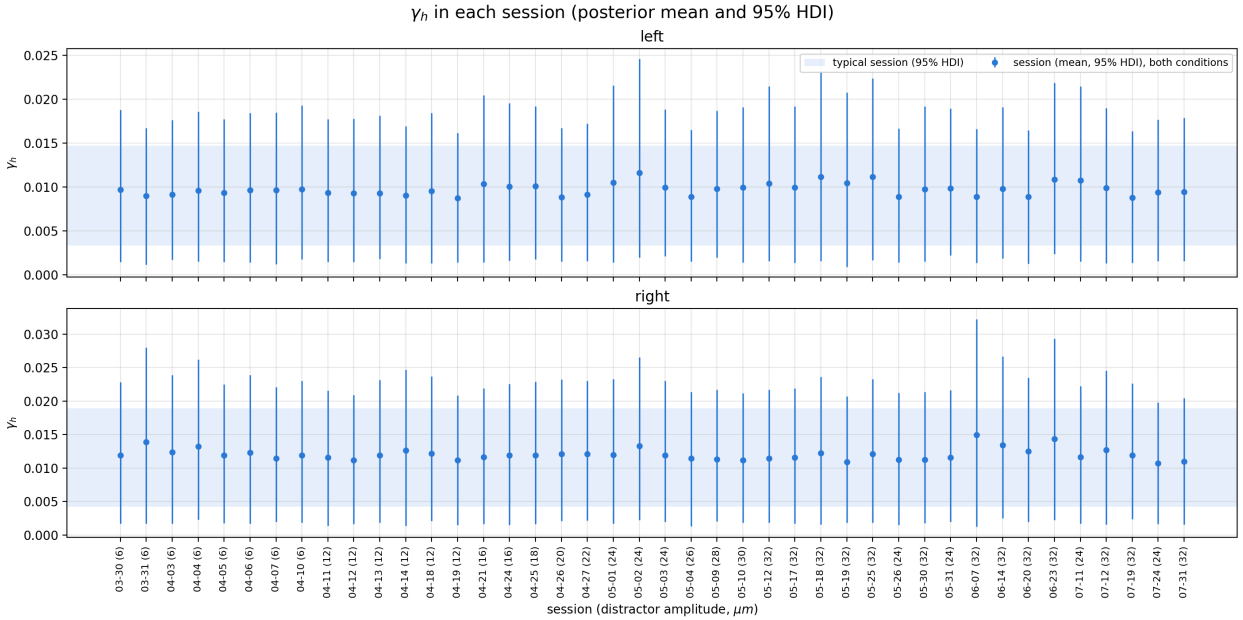


Figure 18: γ_h in each session, shared by both conditions.

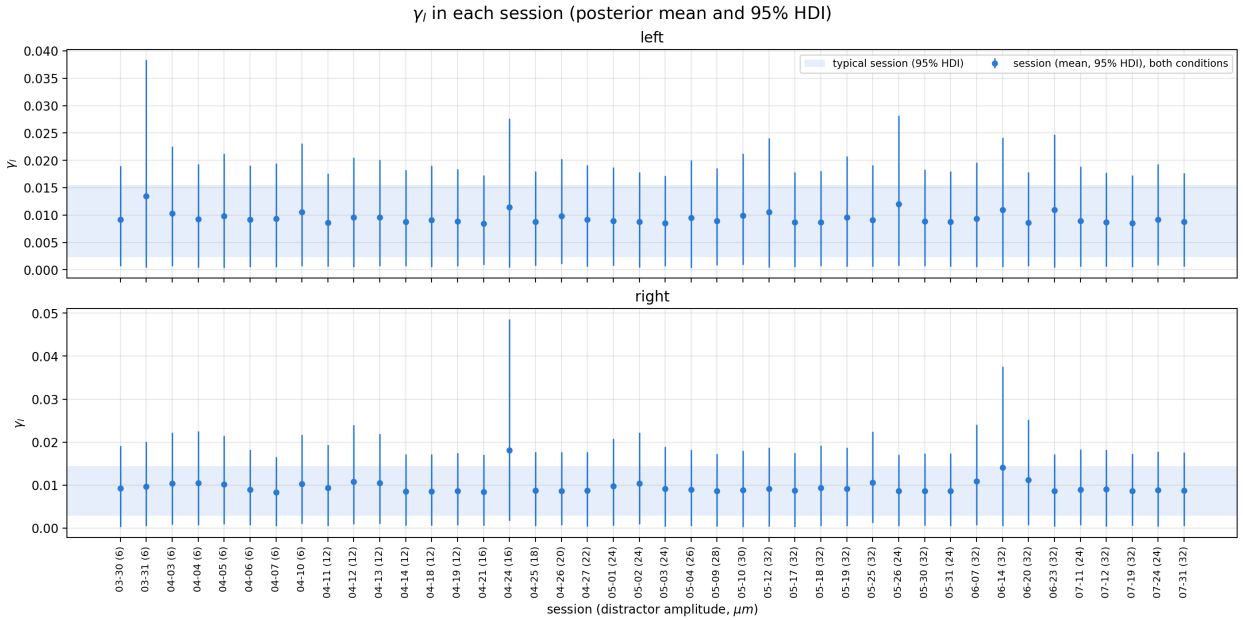


Figure 19: γ_l in each session, shared by both conditions.

The left-hand JNDs vary between sessions ($\sigma_{\beta_1} = 0.90$), from 2.95 to 6.59 μm, with the two shallowest sessions in mid-May; the right-hand ones barely vary, from 3.65 to 4.31 μm (Figure 17). The session lapse rates are almost identical within a hand (Figures 18 and 19): the posterior means of γ_h range only from 0.0087 to 0.0116 (left) and 0.0107 to 0.0149 (right). With σ_{γ_h} and σ_{γ_l} not identified, every session is pulled to its hand's typical value, so these plots say little about real day-to-day differences in lapses.

5 Comparison with Model B

Model B fits unimanual and bimanual trials as separate groups, each with its own per-session parameters, and has no amplitude covariate. Both models were fit to the same trials in the same order.

Leave-one-out cross-validation. Model C has $\mathrm{elpd}_{\mathrm{loo}} = -3217.0$ (SE 65.7) with $p_{\mathrm{loo}} = 138.6$ and every Pareto k at most 0.33; Model B has -3180.0 (SE 64.3) with $p_{\mathrm{loo}} = 178.7$. Model C is worse by 37.1 (SE 11.9), 3.1 standard errors, despite needing fewer effective parameters.

Table 6: $\mathrm{elpd}_{\mathrm{loo}}(\mathrm{C}) - \mathrm{elpd}_{\mathrm{loo}}(\mathrm{B})$ summed over the trials at each amplitude, by hand and condition. Negative values favor Model B.

	left		right	
amplitude (μm)	bimanual	unimanual	bimanual	unimanual
6	−6.6	−3.5	−5.1	−2.3
12	−4.2	−3.0	−1.8	−2.8
18	0.7	−1.6	−3.4	−0.3
24	−1.0	0.8	1.9	0.8
32	−0.1	−3.4	0.3	1.7
38	−0.3	−1.4	0.2	1.6
44	0.0	−3.8	0.4	−1.5
50	−0.1	0.3	0.1	0.0

Model C loses 29.2 of its 37.1 at the 6 and 12 μm stimuli (Table 6): 17.6 on bimanual and 11.6 on unimanual trials. This is the shared-lapse misfit of Section 2.2, seen from both sides. The next largest losses are on left unimanual trials at 32 and 44 μm (-3.4 and -3.8), on the rising part of the curve. With one slope for both conditions, Model C cannot give the left-hand unimanual curve a different steepness than the bimanual one, which Model B allows.

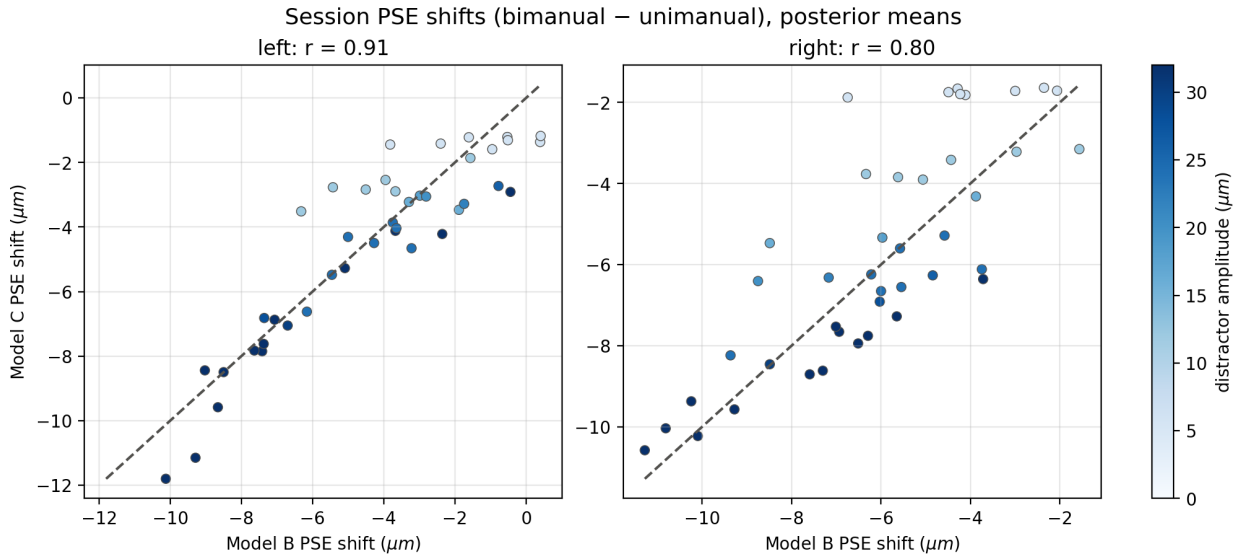


Figure 20: Session PSE shifts (bimanual − unimanual), posterior means: Model C against Model B, colored by the session's distractor amplitude. The dashed line is the identity; r is the correlation.

Session PSE shifts. The two models agree on the average shift (-4.51 against -4.29 μm for the left hand, -5.74 against -6.06 for the right) and, for the left hand, closely on the session shifts ($r = 0.91$, Figure 20). For the right hand ($r = 0.80$) they disagree at the weakest distractor:

Table 7: Session PSE shifts (μm) at the weakest and strongest distractor: mean and range of the posterior means over the sessions with that amplitude.

hand	distractor	Model B	Model C
left	6 μm (8 sessions)	-1.13 (-3.82 to 0.41)	-1.34 (-1.59 to -1.18)
left	32 μm (13 sessions)	-6.67 (-10.12 to -0.44)	-7.40 (-11.80 to -2.91)
right	6 μm (8 sessions)	-3.91 (-6.74 to -2.06)	-1.75 (-1.88 to -1.64)
right	32 μm (13 sessions)	-7.90 (-11.28 to -3.72)	-8.58 (-10.58 to -6.36)

For the right hand, Model B estimates the 6 μm sessions to shift by 3.91 μm on average, Model C by only 1.75 (Table 7). Model C's shift is proportional to the amplitude, so a weak distractor must give a small shift, and with β_2 poorly determined in these sessions they all fall onto the hand-level line. Model B, which estimates each session's shift freely, suggests that for the right hand even a weak distractor shifts the PSE substantially: an effect of the distractor being present, on top of its amplitude.

6 Conclusions

- The fit is reliable: 2 divergences in 20,000 draws, $\hat{R} = 1.000$ and bulk ESS $\geq$ 3,410 for every variable, every Pareto k good.
- The PSE shift grows with distractor amplitude for both hands ($\mu_{\beta_2} > 0$ with posterior probability 1.000), more steeply for the right hand. This is the main new result of Model C, and it supports the exploratory amplitude trend in the Model B report.

- Model C predicts the data worse than Model B (elpd difference −37.1, SE 11.9). The misfit sits where three of its assumptions meet the data:
  1. one lapse rate for both conditions, while the distractor raises γ_h: too few "high" responses are predicted at weak stimuli on bimanual trials, and too many on unimanual ones;
  2. one slope for both conditions, which costs fit on the rising part of the left-hand unimanual curve;
  3. a shift proportional to amplitude: for the right hand at 6 μm, Model C's shift is less than half of Model B's, and the observed bimanual − unimanual difference lies outside Model C's predictive interval (Section 4.1).
- The size of the shift at weak distractors, and Model C's lapse rates and JNDs, should therefore not be taken at face value until these assumptions are relaxed.

Recommended revisions.

1. Lapse rates per hand and condition (unimanual, bimanual). The evidence for this is the strongest.
2. A distractor-presence term, $\beta_3 \cdot \mathbb{1}$[bimanual], so that the PSE shift is $-(\beta_3 + \beta_2 d)/\beta_1$ rather than proportional to d.
3. Possibly a separate slope on bimanual trials, which would let the distractor change the JND.

Each revision changes the model, so it would need a new convergence study before being fit to the real data again.

A Joint Posteriors

`Main_Analysis_C.py` also plots joint posteriors for session 04-13 (Figures 21–23). The new parameter β_2 is negatively correlated with β_0 ($r = -0.35$ left, -0.25 right): a session's bimanual position is $\beta_0 + \beta_2 d_s$, so a higher bias can be traded for a smaller distractor effect, but the unimanual trials pin down β_0 well enough to keep the correlation moderate. All other correlations between the session parameters are at most 0.32 in absolute value (that of β_0 and β_1 for the right hand). There is no ridge or multimodality that would indicate an identifiability problem.

The left hand's γ_h on 04-13 is correlated with its hyperparameter μ_{γ_h} ($r = 0.60$) but hardly with σ_{γ_h} ($r = -0.09$) (Figure 23): with σ_{γ_h} small, the session value follows the hand's typical value.

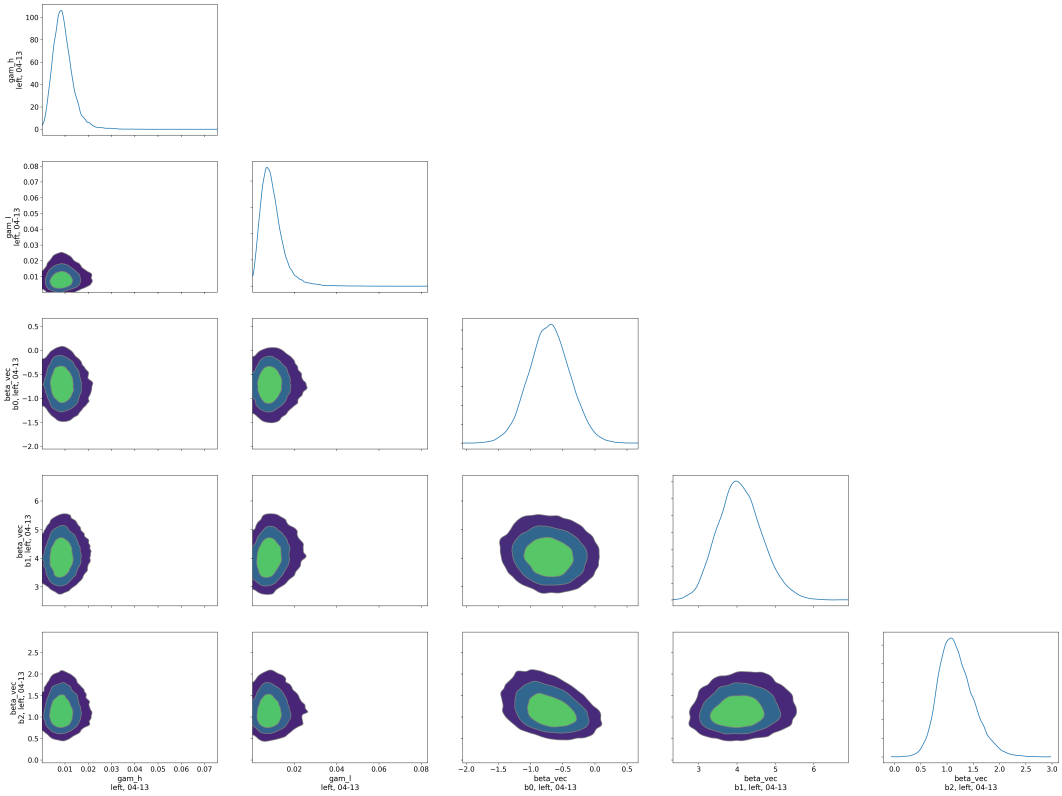


Figure 21: Joint posterior of γ_h, γ_l, β_0, β_1 and β_2 for the left hand, session 04-13.

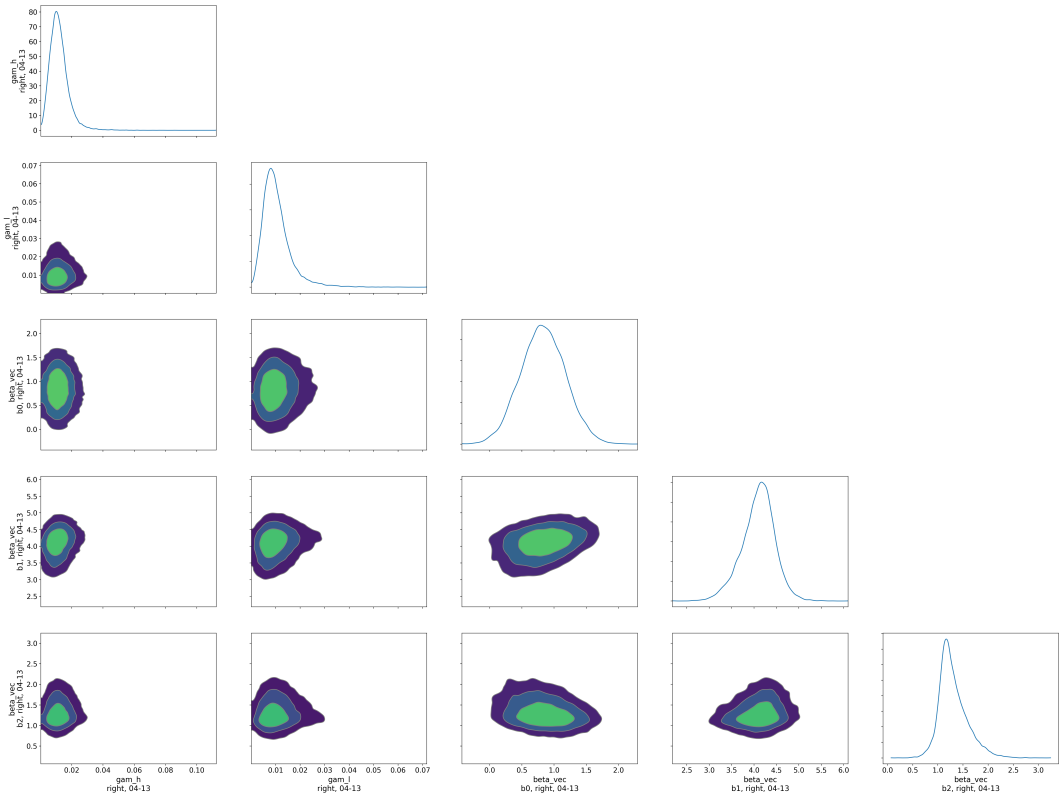


Figure 22: As Figure 21, for the right hand.

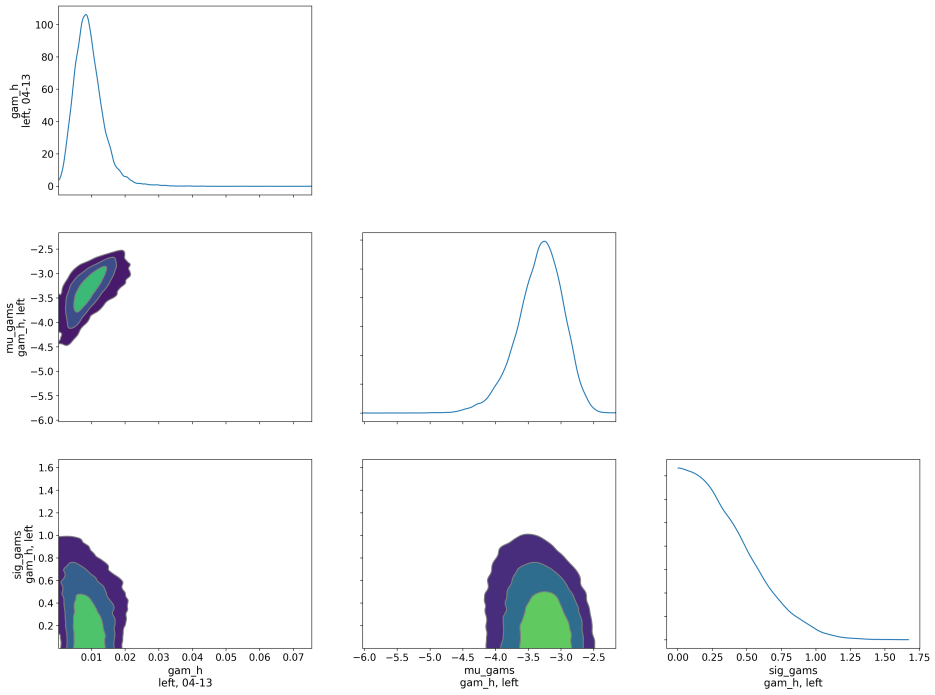


Figure 23: Joint posterior of the left hand's γ_h on 04-13 and the hyperparameters μ_{γ_h} and σ_{γ_h}.